

**Unit ID: 6798bb3eaaf593b2823f123a**

## **Unit 4 - Code Quest**

### **PDF Documents**

#### **PDF Document 1: 1738062652362-Chapter 5 - unit 4 - Code Quest.pdf**

This PDF document is included in the unit materials.

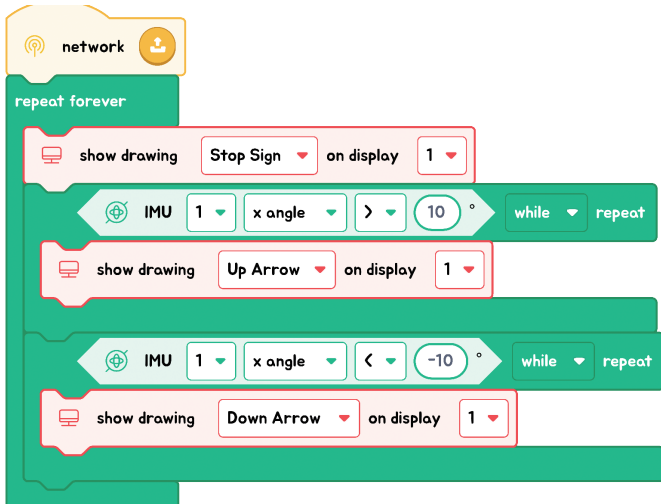
**Update #1 - 2025-05-16 15:17:39**



## UNIT 4 CODE QUEST

### Applying Knowledge (code & debugging)

#### CODE



#### DEBUGGING

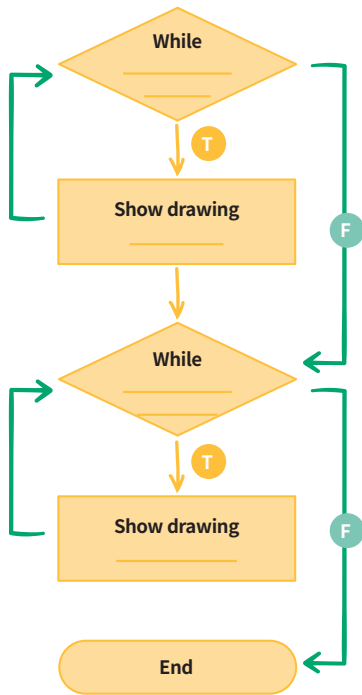
- 1 What if we use 'if statement' instead of 'while statement'?
- 2 Which block do we need to add 'while statement' for stop sign using 'and' operator?



## Missions

### Mission 1

1 Let's add right and left direction at the bottom of original code.



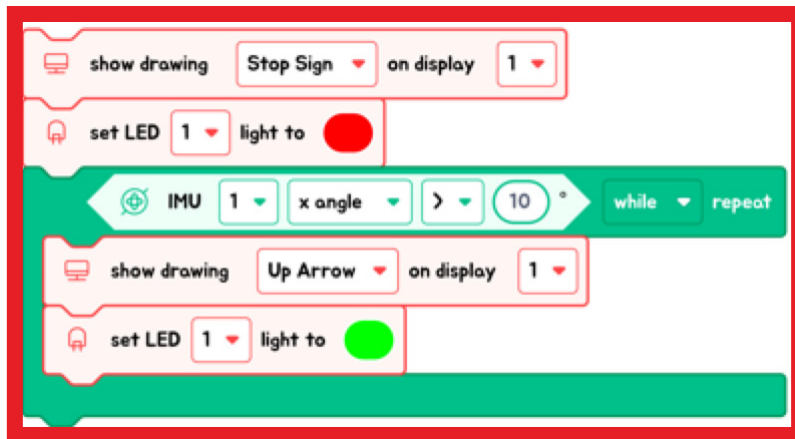
2 Based on the flowchart, make the code

**Misson 2****1** Let's make the LEDs light up depending on the direction

| Direction | Color   |
|-----------|---------|
| Straight  | ex) red |
| Back      |         |
| Right     |         |
| Left      |         |



## 2 Based on the code, make the flowchart.



### YOUR FLOWCHART

Area for drawing the flowchart based on the provided code.

# KNOWLEDGE CHECK

**1** Which direction did the Silk Road connect?

- ☐ A North and South
- ☐ B East and West
- ☐ C East and South
- ☐ D West and South

CORRECT ANSWER

☐ B

**2** How did travelers find their way on the Silk Road?

- ☐ A By following signs
- ☐ B By using GPS
- ☐ C By asking for directions
- ☐ D By using natural landmarks and stars

CORRECT ANSWER

☐ D

**3** What was the Silk Road?

- ☐ A A road made of silk
- ☐ B A historic trade route
- ☐ C A modern highway
- ☐ D A river in China

CORRECT ANSWER

☐ B

**4** What is IMU can not detect?

- ☐ A Vibration
- ☐ B X angle
- ☐ C Y Angle velocity
- ☐ D Temperature

CORRECT ANSWER

☐ D

**5** What is the purpose of a while statement in programming?

- ☐ A To check 2 or more conditions are met
- ☐ B To define a function in the program
- ☐ C To repeat code if a certain condition is true
- ☐ D To declare variables and assign values to them

CORRECT ANSWER

☐ C

**6** Which of the following is NOT available with display module

- ☐ A Write a sentence
- ☐ B Show a picture
- ☐ C Show a number
- ☐ D Draw a picture

CORRECT ANSWER

☐ D

**7** Which of the following is a function of the IMU??

- ☐ A Clicked
- ☐ B Vibration
- ☐ C Turn
- ☐ D Temperature

CORRECT ANSWER

☐ B

**8** Which shape in a flowchart is typically used to represent an while statement?

- ☐ A Rectangle
- ☐ B Diamond
- ☐ C Circle
- ☐ D Parallelogram

CORRECT ANSWER

☐ B



## Critical Thinking Challenge

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**1** What is difference between if statement and while statement?

**2** How Can we make navigation for the blind?